

More about the Team Project

You have located a data set you think you want to use – now what?

Let's assume you are proposing a classification problem. Shown below is a fragment of the data:

Car, V6, low, new, 25, 73, Retail
Car, V8, low, lateModel, 27, 102, Retail
Car, V8, med, lateModel, 21, 57, Retail
Car, V8, high, old, 22, 137, Auction
Truck, Diesel, med, new, 31, 98, Retail
Car, Electric, med, new, 37, 109, Auction
...
...

Here, the attributes are: Car or Truck, engine type, gas efficiency, age of car, initial selling price (rounded to thousands of dollars), mileage (in thousands), **how disposed****.

**The data was originally used as a classification problem, so the class in this case is the right-most attribute (how disposed).

Could we use this data to determine if there are other interesting patterns?

Maybe we could hypothesize that we can determine the age of the car based on the attributes: Car or Truck, engine type, gas efficiency, initial selling price (rounded to thousands of dollars), mileage (in thousands), how disposed. Notice that here we are using “how disposed” as an *attribute* to see if we can classify the age of the car.

As we discussed in class, you may not need all the attributes to classify the age of the car – that would be a part of your analysis. You may also find that you cannot accurately classify the age of the car based on the attributes given. This is what I called a “successful failure.” However, you really want a “success” for the project, so select your data and generate your hypotheses carefully!

I suggest that you come up with at least 2 hypotheses in case you determine that you don't have sufficient data to support a hypothesis.

Avoid hypothesis that will be difficult to support. For example, notice that I did not hypothesize that I could determine “engine type” because the data had too few Diesel and Electric vehicles. Again – this is what we discussed in class so you need to be sure you are selecting a strong hypothesis after taking a careful look at the data.

This is an over-simplified example, but I hope it gives you an idea of where to start.